Crest

Release 4.22.4

Wave Harmonic & Contributors

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CHAPTER

ONE

INTRODUCTION

Crest is a technically advanced ocean system for Unity.

It is architected for performance and makes heavy use of Level Of Detail (LOD) strategies and GPU acceleration for fast update and rendering. It is also highly flexible and allows any custom input to the water shape/foam/dynamic waves/etcetera, and has an intuitive and easy to use shape authoring interface.

This documentation is for *Crest* 4.22.4 and targets URP (Universal Render Pipeline).

You can view this documentation online.

1.1 Assets

You can find Crest 4 on the *Unity Asset Store* for all render pipelines (Crest 5 is also available):

- Crest Ocean System: Asset Store
- Crest Ocean System HDRP: Asset Store
- Crest Ocean System URP: Asset Store

1.2 Social

- **YouTube** https://www.youtube.com/@WaveHarmonic
- **Twitter** https://twitter.com/@crest_ocean
- **Discord** https://discord.gg/JJjx9qcq83

CHAPTER

TWO

INTEGRATIONS

The following is a list of assets that have known integrations for Crest or have general support for transparent objects like water. This list is not exhaustive, and there may be assets that work without declaring support. Furthermore, not all integrations have been tested by us.

Important: For anything render pipeline specific, be sure to check which render pipeline variant of Crest the asset supports.

2.1 Atmosphere

Assets that handle atmosphere (ie sky and weather) often need integrating to apply their fog to transparent objects. The following have provided instructions to integrate with Crest.

- Atmospheric Height Fog: Asset Store
- Azure[Sky]: Asset Store
- Buto: Asset Store
- Cozy 3: Asset Store
- Enviro 3: Asset Store
- Expanse: Asset Store
- Volumetric Fog & Mist 2: Asset Store

2.2 Physics

Crest provides an API for querying the water which can be used for physics like buoyancy and drag.

- Dynamic Water Physics 2: Asset Store

2.3 World Building

World building assets can provide presets to place Crest into the world.

- Gaia Pro: Asset Store
- MicroVerse: Asset Store

2.4 Miscellaneous

Anything that could not be categorized.

- Real-Time Weather Pro: Asset Store

CHAPTER

THREE

KNOWN ISSUES

We keep track of issues on GitHub for all pipelines. Please see the following links:

- Issues on GitHub.
- URP specific issues on GitHub.

If you discover a bug, please open a bug report or mention it on the bugs channel on our Discord.

3.1 Unity Bugs

There are some Unity issues that affect *Crest*. Some of these may even be blocking new features from being developed. A list of these issues can be found on our wiki which can be voted on for Unity to prioritise.

As a reminder it is important to always use the latest patched Unity version.

3.2 Prefab Mode Not Supported

Crest does not support running in prefab mode which means dirty state in prefab mode will not be reflected in the scene view. Save the prefab to see the changes.

CHAPTER

FOUR

RELEASE NOTES

4.22.4

Changed

- Expose Sea Floor Depth input Relative property in the inspector
- Improve platform compatibility by adding more checks to RT formats
- No longer log RT format changes (use CREST_DEBUG_LOG_FORMAT_CHANGES if needed)

Fixed

- Fix Unity 6.4 compilation errors and warnings
- Fix TimeProviderCutscene broken in builds
- Fix water potentially breaking in builds due to RT format
- Fix Examples navigation when only new input system is active
- Fix input module for old input system spamming errors when only new input system is active in Examples scene

Documentation

- Add section dedicated to the Water Body script

4.22.3

Fixed

- Fix underwater alpha channel handling causing problems with certain post-processing settings
- Fix artifacts and/or missing chunks of water on mobile platforms
- Fix exceptions and broken rendering when using Portal mode

4.22.2

Changed

- Add *Follow Horizontal Motion* option to Albedo inputs
- Add *Repair Shaders* button to OceanRenderer
- Warn if primary light is not directional

Fixed

- Fix underwater shader variants potentially being incorrect in builds by making stripping deterministic
- Fix dynamic waves not scaling correctly across LODs
- Fix atmospheric fog causing hard edges where water intersects the scene

Performance

- Reduce underwater shader variants
- Strip underwater debug variants in builds

4.22.1

Changed

- Add game/scene to planar reflection camera name to better distinguish them

Fixed

- Fix script compilation error when Gaia is installed on Unity 6+
- Fix CS0414 script warning on build
- Fix planar reflection "release target texture" exception
- Fix planar reflection capturing inverted geometry in edit mode
- Fix both scene and game view planar reflection camera executing, when only one should
- Fix several cases of "Screen position out of view frustum" when using planar reflections

4.22

Changed

- Add *Gaia Pro VS* integration
- Add Wind Zone support (directional only)
- Add *Global Wind Direction* to WaterRenderer
- Add *Global Wind Turbulence* to WaterRenderer
- Add *Swell* preset to ShapeGerstner
- Add *Enable Render Queue Sorting* option to allow changing the render queue within the transparent pass. This is useful for some third-party integrations
- Add *Far Clip Plane* property to OceanDepthCache
- Add UnderwaterRenderer.AfterCopyMaterial event to allow easy underwater material changes via script
- No longer disable components silently if they fail validation
- Improve shifting origin handling for tiled textures (like foam)
- Add vertical axis support to shifting origin
- Make some more properties public
- Make Ocean-Underwater material the default

Fixed

- Fix "Screen position out of view frustum" with 2D scene view when planar reflections is active
- Fix PSSL shader compilation error for FFT waves
- Fix OceanDepthCache not working after disabling then enabling in builds
- Fix *Hide Depth Cache Cam* not applying immediately
- Fix OceanDepthCache's very short capture range (from 1,000 to 10,000 and now configurable)
- Fix effects being incorrect based on water depth when sea level is not zero
- Fix missing caustics when using shifting origin on mobile devices
- Fix tiled texture quality being dependent on *LOD Resolution*. They are now scaled by LOD scale. This may require changing your scale values for these textures to look as they did before
- Fix Spline.UpdateSpline not working in builds
- Fix wave spline initialization causing inconsistency between editor and builds
- Fix Unity fog when using orthographic projection
- Fix Unity 6.1 "USE_FORWARD_PLUS" shader compiler warning
- Fix errors when inspecting Shader Graph nodes

Full version history has been omitted for brevity. It can be found at Release Notes.

CHAPTER

FIVE

INITIAL SETUP

This section has steps for importing the *Crest* content into a project, and for adding a new ocean surface to a scene.

Warning: When changing Unity versions, setting up a render pipeline or making changes to packages, the project can appear to break. This may manifest as spurious errors in the log, no ocean rendering, magenta materials, scripts unassigned in example scenes, etcetera. Often, restarting the Editor fixes it. Clearing out the *Library* folder can also help to reset the project and clear temporary errors. These issues are not specific to *Crest*, but we note them anyway as we find our users regularly encounter them.

To augment / complement this written documentation we published a video available here:

https://www.youtube.com/watch?v=TpJf13d_-3E

Fig. 5.1: Getting Start with Crest for URP

5.1 Requirements

- Unity: 2020.3 or later
- Shader compilation target 4.5 or above
- Crest does **not** support OpenGL or WebGL backends
- The minimum URP package version is 10.10

5.2 Importing *Crest* files into project

The steps to set up *Crest* in a new or existing project are as follows:

5.2.1 Pipeline Setup

Ensure that URP is setup and functioning, either by setting up a new project using the URP template or by configuring your current project. This is beyond the scope of this documentation so please see the Unity documentation for more information.

Switch to Linear space rendering under *Edit* → *Project Settings* → *Player* → *Other Settings*. If your platform(s) require Gamma space, the material settings will need to be adjusted to compensate. Please see the Unity documentation for more information.

5.2.2 Importing Crest

Import the *Crest* package into project using the *Asset Store* window in the Unity Editor.

The files under Crest-Examples are not required by our core functionality, but are provided for illustrative purposes. We recommend first time users import them as they may provide useful guidance.

Tip: If you are starting from scratch we recommend creating a project using a template in the Unity Hub.

5.3 Adding *Crest* to a Scene

The steps to add an ocean to an existing scene are as follows:

- Create a new *GameObject* for the ocean, give it a descriptive name such as *Ocean*.
  - Assign the *OceanRenderer* component to it. This component will generate the ocean geometry and do all required initialisation.
  - Assign the desired ocean material to the *OceanRenderer* script - this is a material using the *Crest/Ocean* shader.
  - Set the Y coordinate of the position to the desired sea level.
- Tag a primary camera as *MainCamera* if one is not tagged already, or provide the *Camera* to the *View Camera* property on the *OceanRenderer* script. If you need to switch between multiple cameras, update the *ViewCamera* field to ensure the ocean follows the correct view.
- Be sure to generate lighting if necessary. The ocean lighting takes the ambient intensity from the baked spherical harmonics. It can be found at the following:

  Window → *Rendering* → *Lighting Settings* → *Debug Settings* → *Generate Lighting*

  Tip: You can check *Auto Generate* to ensure lighting is always generated.

- To add waves, create a new GameObject and add the *Shape FFT* component. See *Waves* section for customisation.
- Any ocean seabed geometry needs set up to register it with *Crest*. See section *Shorelines and Shallows*.
- If the camera needs to go underwater, the underwater effect must be configured. See section *Underwater* for instructions.

5.4 Frequent Setup Issues

The following are kinks or bugs with the install process which come up frequently.

Errors present, or visual issues

Try restarting Unity as a first step.

Compile errors in the log, not possible to enter play mode, visual issues in the scene

Verify that render pipeline is installed and enabled in the settings. See the follow for documentation:

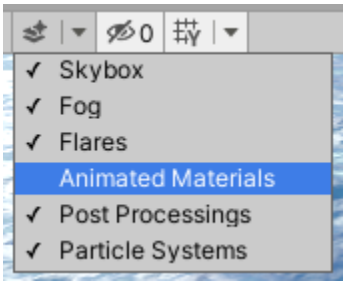
Installing URP into a project

Possible to enter play mode, but errors appear in the log at runtime that mention missing 'kernels'

Recent versions of Unity have a bug that makes shader import unreliable. Please try reimporting the *Crest/Shaders* folder using the right click menu in the project view. Or simply close Unity, delete the Library folder and restart which will trigger everything to reimport.

Ocean framerate low in edit mode

By default, the update speed is intentionally throttled by Unity to save power when in edit mode. To enable real-time update, enable *Animated Materials* in the Scene View toggles:



See the Unity Documentation for more information.

Changes made in prefab mode are not reflected in the scene view

Crest does not support running in prefab mode which means dirty state in prefab mode will not be reflected in the scene view. Save the prefab to see the changes.

CHAPTER

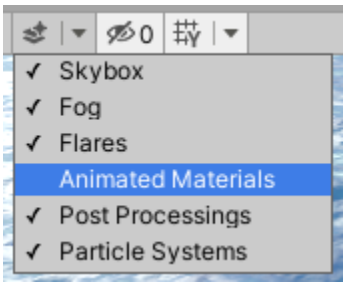
SIX

QUICK START GUIDE

This section provides a summary of common steps for setting up water with links for further reading:

- **Add water**: Add *Crest* water to your scene as described in section *Adding Crest to a Scene*.
- **Ocean surface appearance**: The active ocean material is displayed below the *OceanRenderer* component. The material parameters are described in section *Material Parameters*. Turn off unnecessary features to maximize performance.
- **Add Waves**: Add *Shape FFT* component to a GameObject and assign a *Ocean Wave Spectrum* asset. Waves can be generated everywhere, or in specific areas by placing them on a *Spline*. See section *Waves*.
- **Shallow Water**: To reduce waves in shallow water capture the underlying terrain shape into a *Ocean Depth Cache*. See section *Shorelines and Shallows*.
- **Oceans, Rivers and Lakes**: *Crest* supports setting up networks of connected water bodies, by setting the sea level via *Spline* inputs. See section *Oceans, Rivers and Lakes*.
- **Underwater**: If the camera needs to go underwater, the underwater effect must be enabled. See section *Underwater*.
- **Dynamic wave simulation**: Simulates dynamic effects like object-water interaction. See section *Dynamic Waves*.
- **Boats**: Several components combined can create a convincing boat. See page *Watercraft*.
- **Networking**: *Crest* is built with networking in mind and can synchronise waves across the network. It also has limited support for headless servers. See section *Network Synchronisation*.
- **Open Worlds**: *Crest* comes with 'shifting origin' support to enable large open worlds. See section *Shifting Origin*.

Tip: By default, the update speed is intentionally throttled by Unity to save power when in edit mode. To enable real-time update, enable *Animated Materials* in the Scene View toggles:



See the Unity Documentation for more information.

CHAPTER

SEVEN

WATER APPEARANCE

7.1 Material Parameters

7.1.1 Normals

Overall Normal Strength Strength of the final surface normal (includes both wave normal and normal map)
Use Normal Map Whether to add normal detail from a texture. Can be used to add visual detail to the water surface
Normal Map Normal map and caustics distortion texture (should be set to Normals type in the properties)
Normal Map Scale Scale of normal map texture
Normal Map Strength Strength of normal map influence

7.1.2 Scattering

Scatter Colour Base Base colour when looking straight down into water.
Scatter Colour Grazing Base colour when looking into water at shallow/grazing angle.
Enable Shadowing Changes colour in shadow. Requires 'Create Shadow Data' enabled on OceanRenderer script.
Scatter Colour Shadow Base colour in shadow. Requires 'Create Shadow Data' enabled on OceanRenderer script.

7.1.3 Subsurface Scattering

Enable Whether to to emulate light scattering through the water volume.
SSS Tint Colour tint for primary light contribution.
SSS Intensity Base Amount of primary light contribution that always comes in.
SSS Intensity Sun Primary light contribution in direction of light to emulate light passing through waves.
SSS Sun Falloff Falloff for primary light scattering to affect directionality.

7.1.4 Shallow Scattering

Deprecated

Shallow Scattering will be removed in a future version. A properly tweaked *Depth Fog Density* achieves better results at lower cost. Consider copying over the value from our materials.

The water colour can be varied in shallow water (this requires a depth cache created so that the system knows which areas are shallow, see section *Shorelines and Shallows*).

Enable Enable light scattering in shallow water.

Scatter Colour Shallow Scatter colour used for shallow water.

Scatter Colour Depth Max Maximum water depth that is considered 'shallow', in metres. Water that is deeper than this depth is not affected by shallow colour.

Scatter Colour Depth Falloff Falloff of shallow scattering, which gives control over the appearance of the transition from shallow to deep.

Scatter Colour Shallow Shadow Shallow water colour in shadow (see comment on Shadowing param above).

7.1.5 Reflection Environment

Specular Strength of specular lighting response.

Smoothness Smoothness of surface.

Vary Smoothness Over Distance Helps to spread out specular highlight in mid-to-background. From a theory point of view, models transfer of normal detail to microfacets in BRDF.

Smoothness Far Material smoothness at far distance from camera.

Smoothness Far Distance Definition of far distance.

Smoothness Falloff How smoothness varies between near and far distance.

Softness Acts as mip bias to smooth/blur reflection.

Light Intensity Multiplier Main light intensity multiplier.

Fresnel Power Controls harshness of Fresnel behaviour.

Refractive Index of Air Index of refraction of air. Can be increased to almost 1.333 to increase visibility up through water surface.

Deprecated
The *Refractive Index of Air* property will be removed in a future version.

Refractive Index of Water Index of refraction of water. Typically left at 1.333.

Planar Reflections Dynamically rendered 'reflection plane' style reflections. Requires OceanPlanarReflection script added to main camera.

Planar Reflections Distortion How much the water normal affects the planar reflection.

7.1.6 Procedural Skybox

Enable Enable a simple procedural skybox. Not suitable for realistic reflections, but can be useful to give control over reflection colour - especially in stylized/non realistic applications.
Base Base sky colour.
Towards Sun Colour in sun direction.
Directionality Direction fall off.
Away From Sun Colour away from sun direction.

7.1.7 Foam

Enable Enable foam layer on ocean surface.
Foam Foam texture.
Foam Scale Foam texture scale.
Foam Feather Controls how gradual the transition is from full foam to no foam.
Foam Tint Colour tint for whitecaps / foam on water surface.
Light Scale Scale intensity of lighting.
Shoreline Foam Min Depth Proximity to sea floor where foam starts to get generated.

7.1.8 Foam 3D Lighting

Enable Generates normals for the foam based on foam values/texture and use it for foam lighting.
Foam Normal Strength Strength of the generated normals.
Specular Fall-Off Acts like a gloss parameter for specular response.
Specular Boost Strength of specular response.

7.1.9 Foam Bubbles

Foam Bubbles Color Colour tint bubble foam underneath water surface.
Foam Bubbles Parallax Parallax for underwater bubbles to give feeling of volume.
Foam Bubbles Coverage How much underwater bubble foam is generated.

7.1.10 Transparency

Enable Whether light can pass through the water surface.
Refraction Strength How strongly light is refracted when passing through water surface.
Depth Fog Density Scattering coefficient within water volume, per channel.

7.1.11 Caustics

Enable Approximate rays being focused/defocused on underwater surfaces.
Caustics Caustics texture.
Caustics Scale Caustics texture scale.
Caustics Texture Grey Point The 'mid' value of the caustics texture, around which the caustic texture values are scaled.
Caustics Strength Scaling / intensity.
Caustics Focal Depth The depth at which the caustics are in focus.
Caustics Depth Of Field The range of depths over which the caustics are in focus.
Caustics Distortion Strength How much the caustics texture is distorted.
Caustics Distortion Scale The scale of the distortion pattern used to distort the caustics.

7.1.12 Underwater

Enable Whether the underwater effect is being used. This enables code that shades the surface correctly from underneath.
Cull Mode Ordinarily set this to *Back* to cull back faces, but set to *Off* to make sure both sides of the surface draw if the underwater effect is being used.

7.1.13 Flow

Example

Flow is demonstrated in the *whirlpool* example scene.

Enable Flow is horizontal motion in water. 'Create Flow Sim' must be enabled on the OceanRenderer to generate flow data.

7.2 Lighting

7.2.1 General

As other shaders would, the ocean will get most its lighting from the primary directional light (AKA sun).

Additional lights are supported unless using Forward+.

See *Water Tile Prefab* for advanced configuration.

7.2.2 Reflections

Reflections contribute hugely to the appearance of the ocean. The look of the ocean will dramatically changed based on the reflection environment.

The Index of Refraction setting controls how much reflection contributes for different view angles.

The base reflection comes from a one of these sources:

- Unity's specular cubemap. This is the default and is the same as what is applied to glossy objects in the scene. It will support reflection probes, as long as the probe extents cover the ocean tiles, which enables real-time update of the reflection environment (see Unity documentation for more details).
- Procedural skybox. Developed for stylized games, this is a simple approximation of sky colours that will give soft results.

This base reflection can then be overridden by dynamic planar reflections. This can be used to augment the reflection with 3D objects such as boat or terrain. This can be enabled by applying the *Ocean Planar Reflections* script to the active camera and configuring which layers get reflected (don't include the *Water* layer). This renders every frame by default but can be configured to render less frequently. This only renders one view but also only captures a limited field of view of reflections, and the reflection directions are scaled down to help keep them in this limited view, which can give a different appearance. Furthermore 'planar' means the surface is approximated by a plane which is not the case for wavey ocean, so the effect can break down. This method is good for capturing local objects like boats and etcetera.

A good strategy for debugging the use of Unity's specular cubemap is to put another reflective/glossy object in the scene near the surface, and verify that it is lit and reflects the scene properly. Crest tries to use the same inputs for lighting/reflections, so if it works for a test object it should work for the water surface as well.

See *Water Tile Prefab* for advanced configuration.

7.2.3 Refractions

Refractions sample from the camera's colour texture. Anything rendered in the transparent pass or higher will not be included in refractions.

See *Transparent Object In Front Of Ocean Surface* for issues with Crest and other refractive materials.

7.2.4 Water Tile Prefab

Crest uses *Mesh Renderers* to render chunks of water throughout the scene. *Mesh Renderers* have many settings for configuring how a mesh responds to different lighting components like probes.

The *Water Tile Prefab* allows you to provide a pre-configured *Mesh Renderer* to override *Crest*'s defaults. Some settings cannot be overriden as they make no sense or require support from *Crest*.

To use this feature, create a prefab with a *Mesh Renderer*. The only other requirement is to **not** have a *Water Chunk Renderer* present in the prefab.

Settings which cannot be overriden are:

- Receive Shadows
- Motion Vectors

Other settings may not have any effect depending on the level of support with the render pipeline.

7.3 Foam

7.3.1 Overview

Crest simulates foam getting generated by choppy water (*pinched*) wave crests) and in shallow water to approximate foam from splashes at shoreline. Each update (default is 30 updates per second), the foam values are reduced to model gradual dissipation of foam over time.

To turn on this feature, enable the *Create Foam Sim* option on the *OceanRenderer* script, and ensure the *Enable* option is ticked in the Foam group on the ocean material.

To configure the foam sim, create a *Foam Sim Settings* asset by right clicking the a folder in the *Project* window and selecting *Create/Crest/Foam Sim Settings*, and assigning it to the OceanRenderer component in your scene.

7.3.2 User Inputs

Foam supports *Spline Mode* and *Renderer Mode*.

Crest supports inputing any foam into the system, which can be helpful for fine tuning where foam is placed. To place foam, add some geometry into the world at the area where foam should be added. Then assign the *RegisterFoamInput* script which will tag it for rendering into the shape, and apply a material with a shader of type *Crest/Inputs/Foam/...*. See the *DepositFoamTex* object in the *whirlpool.unity* scene for an example.

The process for adding inputs is demonstrated in this Fig. 8.1.

The following input shaders are provided under *Crest/Inputs/Foam*:

- **Add From Texture** adds foam values read from a user provided texture. Can be useful for placing 'blobs' of foam as desired, or can be moved around at runtime to paint foam into the sim.
- **Add From Vert Colours** can be applied to geometry and uses the red channel of vertex colours to add foam to the sim. Similar in purpose to *Add From Texture*, but can be authored in a modelling workflow instead of requiring at texture.
- **Override Foam** sets the foam to the provided value. Useful for removing foam from unwanted areas.

7.3.3 Simulation Settings

General Settings

- **Foam Fade Rate** - How quickly foam dissipates. Low values mean foam remains on surface for longer. This setting should be balanced with the generation *strength* parameters below.

Wave foam / whitecaps

Crest detects where waves are 'pinched' and deposits foam to approximate whitecaps.

- **Wave Foam Strength** - Scales intensity of foam generated from waves. This setting should be balanced with the *Foam Fade Rate* setting.
- **Wave Foam Coverage** - How much of the waves generate foam. Higher values will lower the threshold for foam generation, giving a larger area.

Shoreline foam

If water depth input is provided to the system (see **Sea Floor Depth** section below), the foam sim can automatically generate foam when water is very shallow, which can approximate accumulation of foam at shorelines.

- **Shoreline Foam Max Depth** - Foam will be generated in water shallower than this depth. Controls how wide the band of foam at the shoreline will be. Note that this is not a distance to shoreline, but a threshold on water depth, so the width of the foam band can vary based on terrain slope. To address this limitation we allow foam to be manually added from geometry or from a texture, see the next section.
- **Shoreline Foam Strength** - Scales intensity of foam generated in shallow water. This setting should be balanced with the *Foam Fade Rate* setting.

Developer Settings

These settings should generally be left unchanged unless one is experiencing issues.

- **Simulation Frequency** - Frequency to run the foam sim, in updates per second. Lower frequencies can be more efficient but may lead to visible jitter. Default is 30 updates per second.

7.4 Shadows

The shadow data consists of two channels. One is for normal shadows (hard shadow term) as would be used to block specular reflection of the light. The other is a much softer shadowing value (soft shadow term) that can approximately variation in light scattering in the water volume.

This data is captured from the shadow maps Unity renders before the transparent pass. These shadow maps are always rendered in front of the viewer. The Shadow LOD Data then reads these shadow maps and copies shadow information into its LOD textures.

To turn on this feature, enable the *Create Shadow Data* option on the *OceanRenderer* script, and ensure the *Shadowing* option is ticked on the ocean material.

To create this setup from scratch, the steps are the following.

1. In the shadow settings of the URP asset, ensure that shadow cascades are enabled. *Crest* requires cascades to be enabled to obtain shadow information.
2. Enable shadowing in Crest. Enable *Create Shadow Data* on the OceanRenderer script.
3. On the same script, assign a *Primary Light* for the shadows. This light needs to have shadows enabled, if not an error will be reported accordingly.
4. If desired the shadow sim can be configured by assigning a *Shadow Sim Settings* asset (*Create/Crest/Shadow Sim Settings*).
5. Enable *Shadowing* on the ocean material to compile in the necessary shader code

The shadow sim can be configured by assigning a Shadow Sim Settings asset to the OceanRenderer script in your scene (*Create/Crest/Shadow Sim Settings*). In particular, the soft shadows are very soft by default, and may not appear for small/thin shadow casters. This can be configured using the *Jitter Diameter Soft* setting.

There will be times when the shadow jitter settings will cause shadows or light to leak. An example of this is when trying to create a dark room during daylight. At the edges of the room the jittering will cause the ocean on the inside of the room (shadowed) to sample outside of the room (not shadowed) resulting in light at the edges. Reducing the *Jitter Diameter Soft* setting can solve this, but we have also provided a *Register Shadow Input* component which can override the shadow data. This component bypasses jittering and gives you full control.

Shadows only supports the *Renderer Mode*.

7.5 Custom Albedo

7.5.1 Overview

The Albedo feature allows a colour layer to be composited on top of the water surface. This is useful for projecting colour onto the surface.

This is somewhat similar to decals, except the colour only affects the water.

Note: HDRP has a Decal Projector feature that works with the water, and the effect is more configurable and may be preferred over this feature. When using this feature be sure to enable Affects Transparent.

URP 2022 has a decal system but it does not support transparent surfaces like water.

There is a *Render Alpha On Surface* component which is an alternative. It behaves similar to a decal projector, but has several issues like z-order issues.

7.5.2 User Inputs

Note: Inputs only execute the first shader pass (pass zero). It is recommended to use unlit shader templates or unlit *Shader Graph* (URP only) if not using one of ours. Shaders provided by *Unity* generally will not work as their primary pass is not zero - even for unlit shaders.

Albedo only supports the *Renderer Mode*.

Any geometry or particle system can add colour to the water. It will be projected from a top down perspective onto the water surface.

To tag GameObjects to render onto the water, attach the *RegisterAlbedoInput* component.

7.6 Orthographic Projection

Crest supports orthographic projection, but it might require some configuration to get a desired appearance.

Crest uses the camera's position for the LOD system which can be awkward for orthographic which uses the size property on the camera. Use the *Viewpoint* property on the *Ocean Renderer* to override the camera's position. Increasing the minimum scale (*Ocean Renderer* → *Detail Params* → *Min Scale*) will help spread the detail out to make the water appear more uniform. Pick a minimum scale that looks best.

For rendering to look correct, the height of the camera needs to be the same as, or higher than, the camera's size value.

Underwater only works with a positive near clip plane.

7.7 Ocean Construction Parameters

There are a small number of parameters that control the construction of the ocean shape and geometry:

- **Lod Data Resolution** - the resolution of the various ocean LOD data including displacement textures, foam data, dynamic wave sims, etc. Sets the 'detail' present in the ocean - larger values give more detail at increased run-time expense.
- **Geometry Down Sample Factor** - geometry density - a value of 2 will generate one vert per 2x2 LOD data texels. A value of 1 means a vert is generated for every LOD data texel. Larger values give lower fidelity surface shape with higher performance.
- **Lod Count** - the number of levels of detail / scales of ocean geometry to generate. The horizontal range of the ocean surface doubles for each added LOD, while GPU processing time increases linearly. It can be useful to select the ocean in the scene view while running in editor to inspect where LODs are present.
- **Max Scale** - the ocean is scaled horizontally with viewer height, to keep the meshing suitable for elevated viewpoints. This sets the maximum the ocean will be scaled if set to a positive value.
- **Min Scale** - this clamps the scale from below, to prevent the ocean scaling down to 0 when the camera approaches the sea level. Low values give lots of detail, but will limit the horizontal extents of the ocean detail. Increasing this value can be a great performance saving for mobile as it will reduce draw calls.

7.8 Advanced Ocean Parameters

These parameters are found on the *Ocean Renderer* under the *Advanced* heading.

- **Surface Self-Intersection Mode** - How Crest should handle self-intersections of the ocean surface caused by choppy waves which can cause a flipped underwater effect. When not using the portals/volumes, this fix is only applied when within 2 metres of the ocean surface. *Automatic* will disable the fix if portals/volumes are used and is the recommended setting.
- **Underwater Cull Limit** - Proportion of visibility below which ocean will be culled underwater. The larger the number, the closer to the camera the ocean tiles will be culled.

CHAPTER

EIGHT

WATER INPUTS

Inputs provides a means for developers to control the various simulations powering *Crest*. The following video covers the basics:

https://www.youtube.com/watch?v=sQIakAjSq4Y

Fig. 8.1: Basics of Adding Ocean Inputs

8.1 Input Modes

A number of components provide multiple authoring modes depending on other attached components.

8.1.1 Spline Mode

If a Spline component is present, then this mode is activated. It takes priority over all other modes.

This mode requires a *Spline* component to be present with at least two spline points added. Help boxes in the Inspector serve to guide/automate this setup.

Once a *Spline* is created, this is used to drive the input. A common use of splines is to set the water level to follow a riverbed using the *RegisterHeightInput* component. A spline may also be used to add waves or flow velocity, if this gives the required level of fidelity. Another typical use case of splines is to add waves aligned to shorelines.

Relevant data components will automatically be added to spline points. For example if the spline is used with a *RegisterFlowInput* component, the *Spline Point Data Flow* component will be added to spline points which can then be used to configure the flow speed.

Any changes to splines via script require calling *Spline.UpdateSpline*. Updating splines has not been optimized for runtime performance so use conservatively.

8.1.2 Renderer Mode

This is the most advanced type of input and allows rendering any geometry/shader into the water system data. One could draw foam directly into the foam data, or inject a flow map baked from an offline sim.

This mode will activate if a *Renderer* is present. It will take priority except when a *Spline* is present.

The geometry can come from a MeshRenderer, or it can come from any Renderer component such as a TrailRenderer, LineRenderer or ParticleSystem. This geometry will be rendered from a orthographic top down perspective to "print" the data onto the water. For simple cases it is recommended to use an upwards facing quad for the best performance.

The *Particle Renderer* example in the *Examples* scene shows a particle system being projected onto the water surface.

Tip: Inputs only execute the first shader pass (pass zero). It is recommended to use unlit shader templates or unlit *Shader Graph* (URP only) if not using one of ours.

The following shaders can be used with any ocean input:

- **Scale By Factor** scales the ocean data between zero and one inclusive. It is multiplicative, which can be inverted, so zero becomes no data and one leaves the data unchanged.

CHAPTER

NINE

WATER EXCLUSION

Also referred to as clipping or masking, features detailed here can either exclude or include the surface and/or volume.

9.1 Clip Surface

https://www.youtube.com/watch?v=jXphUy__J0o

Fig. 9.1: Water Bodies and Surface Clipping

This data drives clipping of the ocean surface (has no effect on the volume), as in carving out holes. It is similar to Unity's Terrain Holes feature. This can be useful for hollow vessels or low terrain that goes below sea level. Data can come from primitives (signed-distance), geometry (convex hulls) or a texture.

To turn on this feature, enable the *Create Clip Surface Data* option on the *OceanRenderer* script, and ensure the *Enable* option is ticked in the *Clip Surface* group on the ocean material.

The data contains 0-1 values. Holes are carved into the surface when the value is greater than 0.5.

9.1.1 Simulation Settings

All of the settings below refer to the *Clip Surface Sim Settings* asset.

- **Render Texture Graphics Format** - The render texture format to use for the clip surface simulation. Consider using higher precision (like *R16_UNorm*) if you are using *Primitive* mode for even more accurate clipping.

9.1.2 User Inputs

The *Register Clip Surface Input* input only supports the modes listed in the *Mode* dropdown.

Primitive Mode

Clip areas can be added using signed-distance primitives which produces accurate clipping and supports overlapping. Add a *RegisterClipSurfaceInput* script to a *GameObject* and set *Mode* to *Primitive*. The position, rotation and dimensions of the primitive is determined by the *Transform*. See the *FloatingOpenContainer* object in the *boat.unity* scene for an example usage.

Geometry Mode

Clip areas can be added by adding geometry that covers the desired hole area to the scene and then assigning the *RegisterClipSurfaceInput* script and setting *Mode* to *Geometry*. See the *RowBoat* object in the *main.unity* scene for an example usage.

To use other available shaders like *ClipSurfaceRemoveArea* or *ClipSurfaceRemoveAreaTexture*: create a material, assign to renderer and disable *Assign Clip Surface Material* option. For the *ClipSurfaceRemoveArea* shaders, the geometry should be added from a top-down perspective and the faces pointing upwards.

The following input shaders are provided under *Crest/Inputs/Clip Surface*:

- **Convex Hull** - Renders geometry into clip surface data taking all dimensions into account. An example use case is rendering the convex hull of a vessel to remove the ocean surface from within it.

  Example

  See the *RowBoat* object in the *main.unity* scene for an example usage.

  Note: Overlapping or adjacent meshes will not work correctly in most cases. There will be cases where one mesh will overwrite another resulting in the ocean surface appearing where it should not. The mesh is rendered from a top-down perspective. The back faces add clip surface data and the front faces remove from it which creates the convex hull. With an overlapping mesh, the front faces of the sides of one mesh will clear the clipping data creating by the other mesh. Overlapping boxes which are not rotated on the X or Z axes will work well whilst spheres will have issues. Consider using *Primitive* mode which supports overlapping.

- **Include Area** - Removes clipping data so the ocean surface renders.
- **Remove Area** - Adds clipping data to remove the ocean surface.
- **Remove Area Texture** - Adds clipping data using a texture to remove the ocean surface.

9.2 Mask Underwater

The *Portals & Volumes* feature can remove both the water surface and the underwater volume. Otherwise, enable/disable the *Underwater Renderer* where needed.

CHAPTER

TEN

WAVES

The *Animated Waves* simulation contains the animated surface shape. This typically contains the waves from shape components, but can also contain waves from the ripple simulation. All waves will eventually be combined into this simulation so the water shader only needs to sample once to animate vertices.

10.1 Environmental Waves

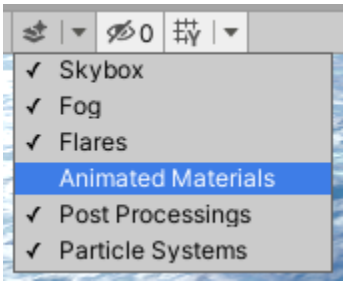
The *ShapeFFT* component is used to generate waves in Crest. To add waves, add the *ShapeFFT* component to a GameObject.

For advanced situations where a high level of control is required over the wave shape, the *ShapeGerstner* component can be used to add specific wave components. It can be especially useful for Trochoidal waves and shoreline waves. See the *Shoreline Waves* section for more information on the latter.

The *Shape Gerstner Batched* component is deprecated.

Tip: It is useful to see the animated ocean surface while tweaking the wave conditions.

By default, the update speed is intentionally throttled by Unity to save power when in edit mode. To enable real-time update, enable *Animated Materials* in the Scene View toggles:



See the Unity Documentation for more information.

10.1.1 Wave Conditions

The appearance and shape of the waves is determined by a *Wave Spectrum*. A default wave spectrum will be created if none is specified. To author wave conditions, click the *Create Asset* button next to the *Spectrum* field. The resulting spectrum can then be edited by expanding this field.

The spectrum can be freely edited in Edit mode, and is locked by default in Play mode to save evaluating the spectrum every frame (this optimisation can be disabled using the *Spectrum Fixed At Runtime* toggle). The spectrum has sliders for each wavelength to control contribution of different scales of waves. To control the contribution of 2m wavelengths, use the slider labelled '2'. Note that the wind speed may need to be increased on the *OceanRenderer* component in order for large wavelengths to be visible.

There is also control over how aligned waves are to the wind direction. This is controlled via the *Wind Turbulence* control on the *ShapeFFT* component.

Another key control is the *Chop* parameter which scales the horizontal displacement. Higher chop gives crisper wave crests but can result in self-intersections or 'inversions' if set too high, so it needs to be balanced.

To aid in tweaking the spectrum, we provide a standard empirical wave spectrum model from the literature, called the 'Pierson-Moskowitz' model. To apply this model to a spectrum, select it in the *Empirical Spectra* section of the spectrum editor which will lock the spectrum to this model. The model can be disabled afterwards which will unlock the spectrum power sliders for hand tweaking.

Tip: Notice how the empirical spectrum places the power slider handles along a line. This is typical of real world wave conditions which will have linear power spectrums on average. However actual conditions can vary significantly based on wind conditions, land masses, etc, and we encourage experimentation to obtain visually interesting wave conditions, or conditions that work best for gameplay.

The waves will be dampened/attenuated in shallow water if a *Sea Floor Depth* LOD data is used (see *Sea Floor Depth*). The amount that waves are attenuated is configurable using the *Attenuation In Shallows* setting.

Together these controls give the flexibility to express the great variation one can observe in real world seascapes.

10.1.2 Wave Placement

Waves can be applied everywhere in the world, placed along or orthogonal to a spline, or injected via a custom shader. See the *Input Modes* section for information about these authoring modes.

Shape components' *Renderer Mode* has custom shaders under *Crest/Inputs/Shape Waves*:

- **Sample Spectrum** samples from the spectrum using a texture. The RG channels are the wave direction and together they make the magnitude. The values are 0-1 where 0.5 is zero magnitude (ie no waves).

10.1.3 Wave Manipulation

The Animated Waves simulation can also be manipulated directly using a *Register Animated Waves Input*.

For the *Register Animated Waves Input*'s *Renderer Mode*, the following shaders are provided with *Crest* under the shader category *Crest/Inputs/Animated Waves*:

- **Scale By Factor** scales the waves by a factor where zero is no waves and one leaves waves unchanged. Useful for reducing waves.
- **Add From Texture** allows any kind of shape added to the surface from a texture. Can ether be a heightmap texture (1 channel) or a 3 channel XYZ displacement texture.

- **Push Water Under Convex Hull** pushes the water underneath the geometry. Can be used to define a volume of space which should stay 'dry'.
- **Wave Particle** is a 'bump' of water. Many bumps can be combined to make interesting effects such as wakes for boats or choppy water. Based loosely on http://www.cemyuksel.com/research/waveparticles/.
- **Set Base Water Height Using Geometry** allows the sea level (average water height) to be offset some amount. The top surface of the geometry will provide the water height, and the waves will apply on top.

  Deprecated

  This shader is deprecated in favour using a *Register Height Input*.

- **Set Water Height Using Geometry** snaps the water surface to the top surface of the geometry. Will override any waves.

  Deprecated

  This shader is deprecated in favour using a *Register Height Input*.

10.1.4 Advanced Settings

The environmental waves are termed *Animated Waves* in the *Crest* system and can be configured by assigning an Animated Waves Sim Settings asset to the OceanRenderer script in your scene (*Create* → *Crest* → *Animated Wave Sim Settings*).

All of the settings below refer to the *Animated Waves Sim Settings* asset.

- **Attenuation In Shallows** - How much waves are dampened in shallow water.
- **Shallows Max Depth** - Any water deeper than this will receive full wave strength. The lower the value, the less effective the depth cache will be at attenuating very large waves. Set to the maximum value (1,000) to disable.
- **Collision Source** - Where to obtain ocean shape on CPU for physics / gameplay.
- **Max Query Count** - Maximum number of wave queries that can be performed when using ComputeShaderQueries.
- **Ping Pong Combine Pass** - Whether to use a graphics shader for combining the wave cascades together. Disabling this uses a compute shader instead which doesn't need to copy back and forth between targets, but it may not work on some GPUs, in particular pre-DX11.3 hardware, which do not support typed UAV loads. The fail behaviour is a flat ocean.
- **Render Texture Graphics Format** - The render texture format to use for the wave simulation. Consider using higher precision (like R32G32B32A32_SFloat) if you see tearing or wierd normals. You may encounter this issue if you use any of the *Set Water Height* inputs.

10.2 Dynamic Waves

10.2.1 Overview

Environmental/animated waves are 'static' in that they are not influenced by objects interacting with the water. 'Dynamic' waves are generated from a multi-resolution simulation that can take such interactions into account.

To turn on this feature, enable the *Create Dynamic Wave Sim* option on the *OceanRenderer* script, and to configure the sim, create or assign a *Dynamic Wave Sim Settings* asset on the *Sim Settings Dynamic Waves* option.

The dynamic wave simulation is added on top of the animated FFT waves to give the final shape.

The dynamic wave simulation is not suitable for use further than approximately 10km from the origin. At this kind of distance the stability of the simulation can be compromised. Use the *ShiftingOrigin* component to avoid travelling far distances from the world origin.

10.2.2 Adding Interaction Forces

Dynamic ripples from interacting objects can be generated by placing one or more spheres under the object to approximate the object's shape. To do so, attach one or more *SphereWaterInteraction* components to children with the object and set the *Radius* parameter to roughly match the shape.

The following settings can be used to customise the interaction:

- **Radius** - The radius of the sphere from which the interaction forces are calculated.
- **Weight** - Strength of the effect. Can be set negative to invert.
- **Weight Up Down Mul** - Multiplier for vertical motion, scales ripples generated from a sphere moving up or down.
- **Inner Sphere Multiplier** - Internally the interaction is modelled by a pair of nested spheres. The forces from the two spheres combine to create the final effect. This parameter scales the effect of the inner sphere and can be tweaked to adjust the shape of the result.
- **Inner Sphere Offset** - This parameter controls the size of the inner sphere and can be tweaked to give further control over the result.
- **Velocity Offset** - Offsets the interaction position in the direction of motion. There is some latency between applying a force to the wave sim and the resulting waves appearing. Applying this offset can help to ensure the waves do not lag behind the sphere.
- **Compensate For Wave Motion** - If set to 0, the input will always be applied at a fixed position before any horizontal displacement from waves. If waves are large then their displacement may cause the interactive waves to drift away from the object. This parameter can be increased to compensate for this displacement and combat this issue. However increasing too far can cause a feedback loop which causes strong 'ring' artifacts to appear in the dynamic waves. This parameter can be tweaked to balance this two effects.

Non-spherical objects can be approximated with multiple spheres, for an example see the *Spinner* object in the *boat.unity* example scene which is composed of multiple sphere interactions. The intensity of the interaction can be scaled using the *Weight* setting. For an example of usages in boats, search for GameObjects with "InteractionSphere" in their name in the *boat.unity* scene.

10.2.3 Simulation Settings

All of the settings below refer to the *Dynamic Wave Sim Settings* asset.

The key settings that impact stability of the simulation are the **Damping** and **Courant Number** settings described below.

- **Simulation Frequency** - Frequency to run the dynamic wave sim, in updates per second. Lower frequencies can be more efficient but may limit wave speed or lead to visible jitter. Default is 60 updates per second.
- **Damping** - How much energy is dissipated each frame. Helps sim stability, but limits how far ripples will propagate. Set this as large as possible/acceptable. Default is 0.05.
- **Courant Number** - Stability control. Lower values means more stable sim, but may slow down some dynamic waves. This value should be set as large as possible until sim instabilities/flickering begin to appear. Default is 0.7.
- **Horiz Displace** - Induce horizontal displacements to sharpen simulated waves.
- **Displace Clamp** - Clamp displacement to help prevent self-intersection in steep waves. Zero means unclamped.
- **Gravity Multiplier** - Multiplier for gravity. More gravity means dynamic waves will travel faster.
- **Attenuation in Shallows** - How much waves are dampened in shallow water.

The *OceanDebugGUI* script gives the debug overlay in the example content scenes and reports the number of sim steps taken each frame.

10.2.4 User Inputs

Dynamic Waves only supports the *Renderer Mode*.

The recommended approach to injecting forces into the dynamic wave simulation is to use the *SphereWaterInteraction* component as described above. This component will compute a robust interaction force between a sphere and the water, and multiple spheres can be composed to model non-spherical shapes.

However for when more control is required custom forces can be injected directly into the simulation using the *Renderer* input mode. The following input shader is provided under *Crest/Inputs/Dynamic Waves*:

- **Add Bump** adds a round force to pull the surface up (or push it down). This can be moved around to create interesting effects.

CHAPTER

ELEVEN

OCEANS, RIVERS AND LAKES

11.1 Oceans

By default Crest generates an infinite body of water at a fixed sea level, suitable for oceans and very large lakes.

11.2 Lakes

Crest can be configured to efficiently generate smaller bodies of water, using the following mechanisms.

- The waves can be generated in a limited area - see the *Spline Mode* section.
- The *WaterBody* component, if present, marks areas of the scene where water should be present. It can be created by attaching this component to a GameObject and setting the X/Z scale to set the size of the water body. If gizmos are enabled an outline showing the size will be drawn in the Scene View.
- The *WaterBody* component turns off tiles that do not overlap the desired area. The *Clip Surface* feature can be used to precisely remove any remaining water outside the intended area. Additionally, the clipping system can be configured to clip everything by default, and then areas can be defined where water should be included. See the *Clip Surface* section.
- If the lake altitude differs from the global sea level, create a spline that covers the area of the lake and attach the *RegisterHeightInput* component which will set the water level to match the spline (or click the *Set Height* button in the *Spline* inspector). It is recommended to cover a larger area than the lake itself, to give a protective margin against LOD effects in the distance.

Example

The *LakesAndRivers.unity* scene contains an example of a lake connected by a river.

Another advantage of the *WaterBody* component is it allows an optional override material to be provided, to change the appearance of the water. Since this feature cannot be applied partially to an ocean tile, and an ocean tile can overlap two water bodies, this feature does not work well with bordering water bodies. If you use this feature and want to still have an ocean, then disable *Water Body Culling* on the *Ocean Renderer*.

11.3 Rivers

Splines can also be used to create rivers, by creating a spline at the water surface of the river, and attaching the following components:

- *RegisterHeightInput* can be used to set the water level to match the spline.
- *RegisterFlowInput* can be used to make the water move along the spline.
- *ShapeFFT* can be used to generate waves that propagate along the river.

The *Add Feature* section of the *Spline* inspector has helper buttons to quickly add these components.

Example

The *LakesAndRivers.unity* scene contains an example of a river connecting two lakes.

11.4 Water Body Component

The Waterbody component can cull water chunks outside its bounds, clip water outside its bounds at the texel level, and override the material of chunks inside its bounds. The typical use case is for creating Closed Lakes, but it is not required.

11.4.1 Clipping

The Water Body component turns off tiles that do not overlap the desired area. The Clip Surface feature can be used to precisely remove any remaining water outside the intended area. Additionally, the clipping system can be configured to clip everything by default, and then areas can be defined where water should be included. See the *Clip Surface* section.

11.4.2 Material Override

The Water Body can override the water material on the water chunks. This can be used to give closed lakes a distinct appearance.

Important: It is important to understand that since this feature cannot be applied partially to a water chunk, and a water chunk can overlap two water bodies, this feature does not work well with bordering water bodies - including bording an ocean. Typically it works best with only a single lake in the scene.

11.4.3 Usage

The Water Body component, if present, marks areas of the scene where water should be present. It can be created by attaching this component to a Game Object and setting the X/Z scale to set the size of the water body. If gizmos are enabled, an outline showing the size will be drawn in the Scene View.

1. Add WaterBody component to a GameObject
2. Position and expand the scale (XZ only) of the GameObject so the bounds covers the desired area (bounds visible with gizmos)
3. Configure clip and material overrides as desired

4. If clipping is enabled, set *Ocean Renderer → Simulation Params → Default Clipping State* to *Everything Clipped*

Tip: If you only want the material override feature, then disable *Ocean Renderer → Ocean Params → Water Body Culling*.

CHAPTER

TWELVE

SHORELINES AND SHALLOWS

Crest requires water depth information to attenuate large waves in shallow water, to generate foam near shorelines, and to provide shallow water shading. The way this information is typically generated is through the *OceanDepthCache* component, which takes one or more layers, and renders everything in those layers (and within its bounds) from a top-down orthographic view to generate a heightfield for the seabed. These layers could contain the render geometry/terrains, or it could be geometry that is placed in a non-rendered layer that serves only to populate the depth cache. By default this generation is done at run-time during startup, but the component exposes other options such as generating offline and saving to an asset, or rendering on demand.

The seabed affects the wave simulation in a physical way - the rule of thumb is *waves will be affected by the seabed when the water depth is less than half of their wavelength*. So for example when the water is 250m deep, this will start to dampen 500m wavelengths from the spectrum, so it is recommended that the seabed drop down to at least 500m away from islands so that there is a smooth transition between shallow and deep water without a 'step' in the sea floor which appears as a discontinuity in the surface waves and/or a line of foam. Alternatively, there is *Shallows Max Depth* on the *Sim Settings Animated Waves* asset which smooths the attenuation to a provided maximum depth where waves will be at full strength.

12.1 Sea Floor Depth

This simulation stores information that can be used to calculate the water depth. Specifically it stores the terrain height, which can then be differenced with the sea level to obtain the water depth. This water depth is useful information to the system; it is used to attenuate large waves in shallow water, to generate foam near shorelines, and to provide shallow water shading. It is calculated by rendering the render geometry in the scene for each LOD from a top down perspective and recording the Y value of the surface.

The following will contribute to ocean depth:

- Objects that have the *RegisterSeaFloorDepthInput* component attached. These objects will render every frame. This is useful for any dynamically moving surfaces that need to generate shoreline foam, etcetera.
- It is also possible to place world space depth caches as described above. The scene objects will be rendered into this cache once, and the results saved. Once the cache is populated it is then copied into the Sea Floor Depth LOD Data. The cache has a gizmo that represents the extents of the cache (white outline) and the near plane of the camera that renders the depth (translucent rectangle). The cache should be placed at sea level and rotated/scaled to encapsulate the terrain.

When the water is e.g. 250m deep, this will start to dampen 500m wavelengths, so it is recommended that the sea floor drop down to around this depth away from islands so that there is a smooth transition between shallow and deep water without a visible boundary.

12.1.1 Setup

https://www.youtube.com/watch?v=jcmqUlboTUk

Fig. 12.1: Depth Cache usage and setup

One way to inform *Crest* of the seabed is to attach the *RegisterSeaFloorDepthInput* component. *Crest* will record the height of these objects every frame, so they can be dynamic.

The *main.unity* example scene has an example of a cache set up around the island. The cache GameObject is called *IslandDepthCache* and has a *OceanDepthCache* component attached. The following are the key points of its configuration:

- The transform position X and Z are centered over the island
- The transform position y value is set to the sea level
- The transform scale is set to 540 which sets the size of the cache. If gizmos are visible and the cache is selected, the area is demarcated with a white rectangle.
- The *Camera Max Terrain Height* is the max height of any surfaces above the sea level that will render into the cache. If gizmos are visible and the cache is selected, this cutoff is visualised as a translucent gray rectangle.
- The *Layers* field contains the layer that the island is assigned to (*Terrain* in our project). Only objects in these layer(s) will render into the cache.
- Both the transform scale (white rectangle) and the *Layers* property determine what will be rendered into the cache.

By default the cache is populated in the *Start()* function. It can instead be configured to populate from script by setting the *Refresh Mode* to *On Demand* and calling the *PopulateCache()* method on the component from script.

Once populated the cache contents can be saved to disk by clicking the *Save cache to file* button that will appear in the Inspector in play mode. Once saved, the *Type* field can be set to *Baked* and the saved data can be assigned to the *Saved Cache* field.

12.1.2 Shoreline Foam

Once the Sea Floor Depth is running, shoreline foam can be configured. See *Shoreline foam* section for more information.

12.1.3 Troubleshooting

Crest runs validation on the depth caches - look for warnings/errors in the Inspector, and in the log at run-time, where many issues will be highlighted. To run validation, click the *Validate Setup* button at the bottom of the *OceanRenderer* component inspector.

To inspect the contents of the cache, look for a child GameObject parented below the cache with the name prefix *Draw_*. It will have a material with a *Texture* property. By double clicking the icon to the right of this field, one can inspect the contents of the cache. The cache will appear black for dry land and red for water that is at least 1m deep.

12.2 Shoreline Waves

Modelling realistic shoreline waves efficiently is a challenging open problem. We discuss further and make suggestions on how to set up shorelines using global waves with *Crest* in the following video.

https://www.youtube.com/watch?v=Y7ny8pKzWMk

Fig. 12.2: Tweaking Shorelines

Alternatively, using *ShapeGerstner* with a spline is an effective way to create shoreline waves. You will need to set *Reverse Wave Weight* to zero to avoid waves also going in the opposite direction and set *Blend Mode* to *Blend* which effectively overwrites existing waves to prevent global waves from interferring.

CHAPTER

THIRTEEN

TIDES AND CURRENTS

13.1 Flow

13.1.1 Overview

Flow is the horizontal motion of the water volumes. It does not affect wave directions, but transports the waves horizontally. This horizontal motion also affects physics.

This can be used to simulate water currents and other waterflow.

Example

See the *whirlpool.unity* example scene where flow is used to rotate the waves and foam around the vortex.

13.1.2 User Inputs

Foam supports *Spline Mode* and *Renderer Mode*.

Crest supports adding any flow velocities to the system. To add flow, add some geometry into the world which when rendered from a top down perspective will draw the desired displacements. Then assign the *RegisterFlowInput* script which will tag it for rendering into the flow, and apply a material using one of the following shaders.

The following input shaders are provided under *Crest/Inputs/Flow*:

The *Crest/Inputs/Flow/Add Flow Map* shader writes a flow texture into the system. It assumes the x component of the flow velocity is packed into 0-1 range in the red channel, and the z component of the velocity is packed into 0-1 range in the green channel. The shader reads the values, subtracts 0.5, and multiplies them by the provided scale value on the shader. The process of adding ocean inputs is demonstrated in Fig. 8.1.

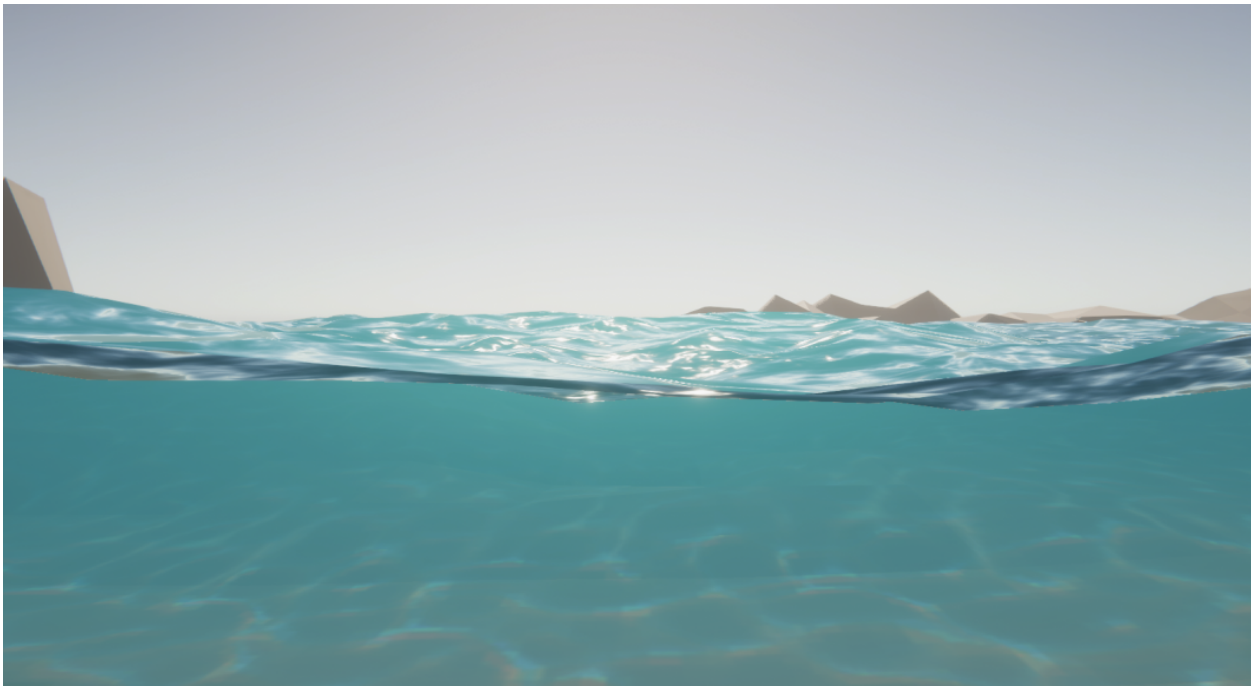
13.2 Tides

It is possible to move the entire water surface on the Y axis to simulate tides.

CHAPTER

FOURTEEN

UNDERWATER



Crest supports seamless transitions above/below water. It can also have a meniscus which renders a subtle line at the intersection between the camera lens and the water to visually help the transition. This is demonstrated in the *main.unity* scene in the example content. The ocean in this scene uses the material *Ocean-Underwater.mat* which enables rendering the underside of the surface.

Out-scattering is provided as an example script which reduces environmental lighting with depth underwater. See the *UnderwaterEnvironmentalLighting* component.

For performance reasons, the underwater effect is disabled if the viewpoint is not underwater. Only the camera rendering the ocean surface will be used.

Tip: Use opaque or alpha test materials for underwater surfaces. Transparent materials may not render correctly underwater. See *Transparent Object Underwater* for possible workarounds.

14.1 Underwater Renderer

Note: You can enable/disable rendering in the scene view by toggling fog in the scene view control bar.

The *Underwater Renderer* component executes a fullscreen underwater effect between the transparent pass and post-processing pass.

It is similar to a post-processing effect, but has the benefit of allowing other renderers to execute after it and still receive post-processing. An example is to add underwater fog correctly to semi-transparent objects.

This is the current underwater solution used for the example scenes, and is the simplest to setup.

14.1.1 Setup

- Configure the ocean material for underwater rendering. In the *Underwater* section of the material params, ensure *Enabled* is turned on and *Cull Mode* is set to *Off* so that the underside of the ocean surface renders. See *Ocean-Underwater.mat* for an example.
- Add the *Underwater Renderer* component to your camera game object.

14.1.2 Parameters

- **Mode:** How the underwater effect (and ocean surface) is rendered:
  - **Full-Screen:** Full screen effect.
  - **Portal:** Renders the underwater effect and ocean surface from the geometry's front-face and behind it.
  - **Volume:** Renders the underwater effect and ocean surface from the geometry's front-face to its back-face.
  - **Volume (Fly-Through):** Renders the underwater effect and ocean surface from the geometry's front-face to its back-face - even from within the volume.
- **Depth Fog Density Factor:** Reduces the underwater depth fog density by a factor. Useful to reduce the intensity of the fog independently from the ocean surface.

14.1.3 Detecting Above or Below Water

The *OceanRenderer* component has the *ViewerHeightAboveWater* property which can be accessed with `OceanRenderer.Instance.ViewerHeightAboveWater`. It will return the signed height from the ocean surface of the camera rendering the ocean. Internally this uses the *SampleHeightHelper* class which can be found in *SamplingHelpers.cs*.

There is also the *OceanSampleHeightEvents* example component (requires example content to be imported) which uses UnityEvents to provide a scriptless approach to triggering changes. Simply attach it to a game object, and it will invoke a UnityEvent when the attached game object is above or below the ocean surface once per state change. A common use case is to use it to trigger different audio when above or below the surface.

14.1.4 Portals & Volumes

The underwater effect can be rendered from a provided mesh which will effectively become a portal (2D) or volume (3D). Change the *Mode* property to one of your choosing and set the *Volume Geometry* to a *Mesh Filter* (it will use its transform). This feature also clips the ocean surface to match. A common use case would be a window on a boat.

14.1.5 Underwater Shader API

The underwater effect uses opaque depth and thus will not render correctly for transparent objects. Too much fog will be applied as it is as if the transparent object does not exist.

The most effective approach is to render the transparent objects after the underwater effect and apply the underwater effect as part of the shader for the transparent object (basically the same way Unity fog is applied).

The *Shader API* needs to be enabled on the *Underwater Renderer* (located under the *Shader API* heading).

Once the *Shader API* is enabled, any transparent object in the correct layer and using a modified shader (more on that later) will have its above water pixels rendered in the transparent pass and below water pixels rendered after the underwater pass.

In a perfect world, we would render the underwater pass before the transparent pass, and then apply the underwater effect to the final color of each transparent object using the *CrestNodeApplyUnderWaterFog* node. But *Shader Graph* does not allow modification of the final color.

The workaround is in the example node *CrestNodeApplyUnderwaterFogExample*. This node uses the *CrestNodeApplyUnderWaterFog* node and does a few things to get around this problem:

- Apply the underwater effect only to the Emission input to bypass *Unity*'s lighting
- Reduce the alpha and the color by distance from the camera

The end result is that the effect is inconsistent with the underwater pass. Despite that we believe it is a decent enough approximation until *Unity* improves this area.

Example

We have an example *Surface Shader* which you can use as a reference:
Crest/Crest-Examples/Shared/Shaders/LitTransparentWithUnderwaterFog.shadergraph

Furthermore, you can view the shader in action in the *Transparent Object Underwater* example in the *Examples* scene.

Setting up a graph can be broken down to the following:

1. Add optional keywords (see example graph)
2. Add the *CrestNodeApplyUnderwaterFogExample* node
3. Connect *Fogged Color* (and alpha) and *Fogged Emission* outputs to the Master Stack
4. Multiply *Factor* output with any properties except *Ambient Occlusion*
5. Enable *Alpha Clipping*

14.2 Underwater Curtain

Deprecated

The *Underwater Curtain* will be removed in a future Crest version. It has been replaced by the *Underwater Renderer*.

14.2.1 Setup

- Configure the ocean material for underwater rendering. In the *Underwater* section of the material params, ensure *Enabled* is turned on and *Cull Mode* is set to *Off* so that the underside of the ocean surface renders. See *Ocean-Underwater.mat* for an example.
- Place *UnderWaterCurtainGeom* and *UnderWaterMeniscus* prefabs under the camera (with cleared transform).

CHAPTER

FIFTEEN

COLLISION SHAPE AND BUOYANCY PHYSICS

15.1 Collision Shape

The system has a few paths for computing information about the water surface such as height, displacement, flow and surface velocity. These paths are covered in the following subsections, and are configured on the *Animated Waves Sim Settings*, assigned to the OceanRenderer script, using the Collision Source dropdown.

The system supports sampling collision at different resolutions. The query functions have a parameter *Min Spatial Length* which is used to indicate how much detail is desired. Wavelengths smaller than half of this min spatial length will be excluded from consideration.

To simplify the code required to get the ocean height or other data from C#, two helpers are provided, *SampleHeightHelper* and *SampleFlowHelper*. Use of these is demonstrated in the example content.

Research

We use a technique called *Fixed Point Iteration* to calculate the water height. We gave a talk at GDC about this technique which may be useful to learn more: https://www.gdcvault.com/play/1023011/Fixed-Point-Iteration-A-Simple.

The *Visualise Collision Area* debug component is useful for visualising the collision shape for comparison against the render surface. It draws debug line crosses in the Scene View around the position of the component.

15.1.1 Collision API Usage

The collision providers built intout our system perform queries asynchronously; queries are offloaded to the GPU or to spare CPU cores for processing. This has a few non-trivial impacts on how the query API must be used.

Firstly, queries need to be registered with an ID so that the results can be tracked and retrieved later. This ID needs to be globally unique, and therefore should be acquired by calling *GetHashCode()* on an object/component which will be guaranteed to be unique. A primary reason why *SampleHeightHelper* is useful is that it is an object in itself and there can pass its own ID, hiding this complexity from the user.

Important: Queries should only be made once per frame from an owner - querying a second time using the same ID will stomp over the last query points.

Secondly, even if only a one-time query of the height is needed, the query function should be called every frame until it indicates that the results were successfully retrieved. See *SampleHeightHelper* and its usages in the code - its *Sample()* function should be called until it returns true. Posting the query and polling for its result are done through the same function.

Finally due to the above properties, the number of query points posted from a particular owner should be kept consistent across frames. The helper classes always submit a fixed number of points this frame, so satisfy this criteria.

15.1.2 Compute Shape Queries (GPU)

This is the default and recommended choice for when a GPU is present. Query positions are uploaded to a compute shader which then samples the ocean data and returns the desired results. The result of the query accurately tracks the height of the surface, including all wave components and depth caches and other Crest features.

15.1.3 Baked FFT Data (CPU)

In scenarios where a GPU is not present such as for headless servers, a CPU option is available.

To use this feature, select a *Shape FFT* component that is generating the waves in a scene and enable the **Enable Baked Collision**. Next configure the following options:

- **Time Resolution** - Frames per second of baked data. Larger values may help the collision track the surface closely at the cost of more frames and increase baked data size.
- **Smallest Wavelength Required** - Smallest wavelength required in collision. To preview the effect of this, disable power sliders in spectrum for smaller values than this number. Smaller values require more resolution and increase baked data size.
- **Time Loop Length** - FFT waves will loop with a period of this many seconds. Smaller values decrease data size but can make waves visibly repetitive.

Next click **Bake to asset and assign to current settings** and select a path and filename for the result. After the bake completes the current active *Animated Waves Sim Settings* will be configured to use this data.

Important: There are currently a few key limitations of this approach:

- Only a single set of waves from one *Shape FFT* component is supported. This collision does not support multiple sets of waves.
- The *Depth Cache* components are not supported. In order to get a one to one match between the visuals and the collision data, depth caches should not be used.
- Varying water levels such as rivers flowing down a gradient or lakes at different altitudes is not supported. This feature assumes a fixed sea level for the whole scene.

15.1.4 Gerstner Waves CPU

Deprecated

The *Shape Gerstner Batched* component is deprecated which supported this option.

This collision option is serviced directly by the *Shape Gerstner Batched* component which implements the *ICollProvider* interface, check this interface to see functionality. This sums over all waves to compute displacements, normals, velocities, etc. In contrast to the displacement textures the horizontal range of this collision source is unlimited.

A drawback of this approach is the CPU performance cost of evaluating the waves. It also does not include wave attenuation from water depth or any custom rendered shape. A final limitation is the current system finds the first

Shape Gerstner Batched component in the scene which may or may not be the correct one. The system does not support cross blending of multiple scripts.

15.2 Buoyancy

Note: Buoyancy physics for boats is not a core focus of *Crest*. For a professional physics solution we recommend the DWP2 (Dynamic Water Physics 2) asset which is compatible with *Crest*.

With that said, we do provide rudimentary physics scripts.

SimpleFloatingObject is a simple buoyancy script that attempts to match the object position and rotation with the surface height and normal. This can work well enough for small water craft that don't need perfect floating behaviour, or floating objects such as buoys, barrels, etc.

BoatProbes is a more advanced implementation that computes buoyancy forces at a number of *ForcePoints* and uses these to apply force and torque to the object. This gives more accurate results at the cost of more queries.

BoatAlignNormal is a rudimentary boat physics emulator that attaches an engine and rudder to *SimpleFloatingObject*. It is not recommended for cases where high animation quality is required.

CHAPTER

SIXTEEN

TIME CONTROL

By default, *Crest* uses the current game time given by Time.time when simulating and rendering the water. In some situations it is useful to control this time, such as an in-game pause or to synchronise wave conditions over a network. This is achieved through what we call *TimeProviders*, and a few use cases are described below.

Note: The *Dynamic Waves* simulation must progress frame by frame and can not be set to use a specific time, and also cannot be synchronised accurately over a network.

16.1 Supporting Pause

One way to pause time is to set Time.timeScale to 0. In many cases it is desirable to leave Time.timeScale untouched so that animations continue to play, and instead pause only the water. To achieve this, attach a *TimeProviderCustom* component to a GameObject and assign it to the *Time Provider* parameter on the *OceanRenderer* component. Then time can be paused by setting the *_paused* variable on the *TimeProviderCustom* component to *false*.

The *TimeProviderCustom* also allows driving any time to the system which may give more flexibility for specific use cases.

A final alternative option is to create a new class that implements the *ITimeProvider* interface and call *OceanRenderer.Instance.PushTimeProvider()* to apply it to the system.

16.2 Network Synchronisation

A requirement in networked games is to have a common sense of time across all clients. This can be specified using an offset between the clients Time.time and that of a server.

This is supported by attaching a *TimeProviderNetworked.cs* component to a GameObject, assigning it to the *Time Provider* parameter on the *OceanRenderer* component, and at run-time setting *TimeProviderNetworked.TimeOffsetToServer* to the time difference between the client and the server.

If using the Mirror network system, set this property to the network time offset. *Crest* expects that if the client time is ahead then the offset needs to be a negative value which may mean that you need to invert the sign of the offset first.

If the server needs the water shape to run physics but does not have a GPU then we have a CPU path, see *Baked FFT Data (CPU)*. Different server conditions can be emulated in Editor using the *Force Batch Mode* and *Force No GPU* toggles on the *OceanRenderer*.

Note that dynamic waves are not synchronised across the network and should not be relied upon in multiplayer projects.

16.3 Timelines and Cutscenes

One use case for this is for cutscenes/timelines when the waves conditions must be known in advance and repeatable. For this case you may attach a *Cutscene Time Provider* component to a GameObject and assign it to the *Ocean Renderer* component. This component will take the time from a Playable Director component which plays a cutscene Timeline. Alternatively, a *Time Provider Custom* component can be used to feed any time into the system, and this time value can be keyframed, giving complete control over timing.

CHAPTER

SEVENTEEN

OPEN WORLDS

Crest follows the camera so it is inheritently suitable for open worlds, but there will be engine issues to account for.

17.1 Shifting Origin

Crest has support for 'shifting origin' functionality, based on code from the *Unity Community Wiki* which is no longer available.

By default the *ShiftingOrigin* script will call *FindObjectsOfType()* for a few different component types, which is a notoriously expensive operation. It is possible to provide custom lists of components to the "override" fields, either by hand or programmatically, to avoid searching the entire scene(s) for the components. Managing these lists at run-time is left to the user.

17.1.1 Stable World Shifts

Tip: Not changing the threshold default value is the simplest approach to stable world shifts.

To avoid popping in the waves, the threshold needs to be set to *WaveResolution x LargestWavelength*. The *WaveResolution* is on the Shape component (*Shape FFT* or *Shape Gerstner*) as just *Resolution*. The *LargestWaveLength* can be found on the *Ocean Wave Spectrum* (the sliders with numbers from 0.0625 to 512) - find the largest number you are using.

For example, a *Shape FFT* with a resolution of 16 and a largest wavelength of 256 will require a threshold value at a minimum of 4,096 to have stable shifts. You could halve this number once at the cost of some instability, but anything low will have very noticeable shifts.

CHAPTER

EIGHTEEN

WATERCRAFT

Note: Buoyancy physics for boats is not a core focus of *Crest*. For a professional physics solution we recommend the DWP2 asset which is compatible with *Crest*.

18.1 Boats

18.1.1 Adding Boats

This section describes the simplest way to add a boat, including buoyancy and wakes, to your project.

Setting up a boat with physics can be a dark art. The authors recommend duplicating and modifying one of the existing boat prefabs, and proceeding slowly and carefully as follows:

1. Pick an existing boat to replace. Only use *BoatAlignNormal* if good floating behaviour is not important, as mentioned above. The best choice is usually *BoatProbes*.
2. Duplicate the prefab of the one you want to replace, such as *crest/Assets/Crest/Crest-Examples/BoatDev/Data/BoatProbes.prefab*
3. Remove the render meshes from the prefab, and add the render mesh for your boat. We recommend lining up the meshes roughly.
4. Switch out the collision shape as desired. Some people report issues if the are multiple overlapping physics collision primitives (or multiple rigidbodies which should never be the case). We recommend keeping things as simple as possible and using only one collider if possible.
5. We recommend placing the render mesh so its approximate center of mass matches the center of the collider and is at the center of the boat transform. Put differently, we usually try to eliminate complex hierarchies or having nested non-zero'd transforms whenever possible within the boat hierarchy, at least on or above physical parts.
6. If you have followed these steps you will have a new boat visual mesh and collider, with the old rigidbody and boat script. You can then modify the physics settings to move the behaviour towards how you want it to be.
7. The mass and drag settings on the boat scripts and rigdibody help to give a feeling of weight.
8. Set the boat dimension:
   - BoatProbes: Set the *Min Spatial Length* param to the width of the boat.
   - BoatAlignNormal: Set the *Boat Width* and *Boat Length* params to the width and length of the boat.
   - If, even after experimenting with the mass and drag, the boat is responding too much to small waves, increase these parameters (try doubling or quadrupling at first and then compensate).
9. There are power settings for engine turning which also help to give a feeling of weight.

10. The dynamic wave interaction is driven by the object in the boat hierarchy called *SphereWaterInteraction*. It can be scaled to match the dimensions of the boat. The *Weight* param controls the strength of the interaction.

The above steps should maintain a working boat throughout - we recommend testing after each step to catch issues early.

18.1.2 Adding Buoyancy

The simplest method to adding buoyancy is detailed above. Further details about buoyancy components can be found in the *Buoyancy* section.

18.1.3 Adding Wakes

The *Sphere Water Interaction* component is used to add wakes. See *Adding Interaction Forces* section for more information on this component.

18.1.4 Removing Water From Inside Boat

There are various methods to removing water from Crest detailed on the *Water Exclusion* page.

CHAPTER

NINETEEN

PERFORMANCE GUIDE

The foundation of *Crest* is architected for performance from the ground up with an innovative LOD system. It is tweaked to achieve a good balance between quality and performance in the general case, but getting the most out of the system requires tweaking the parameters for the particular use case. These are documented below.

19.1 Quality parameters

These are available for tweaking out of the box and should be explored on every project:

- See *Ocean Construction Parameters* for parameters that directly control how much detail is in the ocean, and therefore the work required to update and render it. These are the primary quality settings from a performance point of view.
- The ocean shader has accrued a number of features and has become a reasonably heavy shader. Where possible these are on toggles and can be disabled, which will help the rendering cost (see *Material Parameters*). A potential idea would be to change materials on the fly from script, for example to switch to a deep water material when out at sea to avoid shallow water calculations
- Our wave system uses an inefficient approach to generate the waves to avoid an incompatibility in older hardware. If you are shipping on a limited set of hardware which you can test the waves on, you may try disabling the *Ping pong combine* option in the *Animated Wave Settings* asset.

19.2 Mobile Performance

Mobile is not the primary target for *Crest*, but the following are some hints on getting better performance:

- *Crest* can be draw call heavy which mobile platforms can be sensitive to. Together, reducing the LOD count and increasing the Min Scale can significantly reduce draw calls.
- The *Underwater Renderer* can be very expensive as it will render the water mesh a second time to create a mask. Even though it has been deprecated, try using the *Underwater Curtain* instead.
- Disabling *Transparency* on the ocean material will benefit platforms that use tile-based renderers.

CHAPTER

TWENTY

SYSTEM NOTES

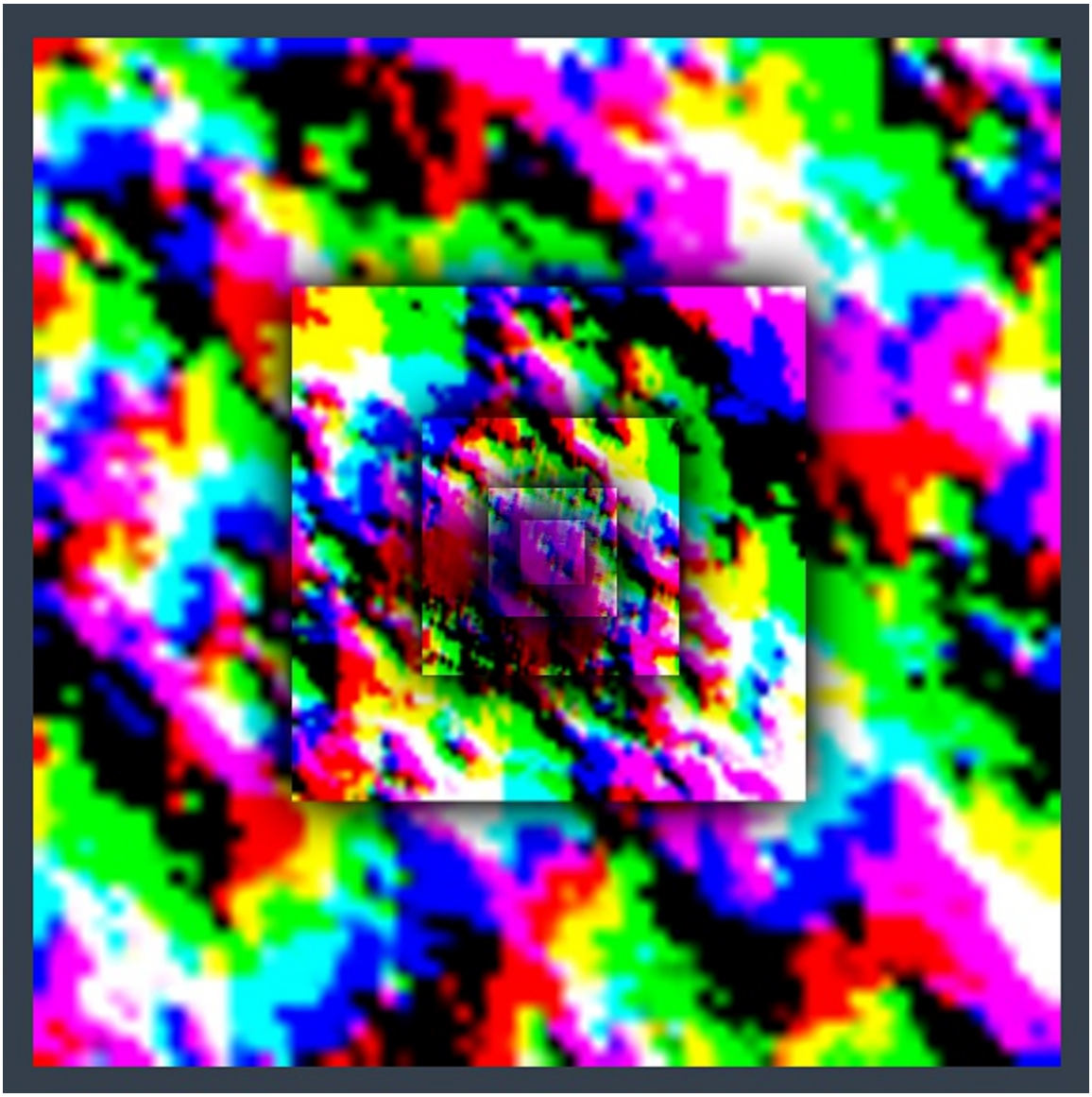
We have published details of the algorithms and approaches we use. See the following publications:

- Crest: *Novel Ocean Rendering Techniques in an Open Source Framework*, Advances in Real-Time Rendering in Games, ACM SIGGRAPH 2017 courses http://advances.realtimerendering.com/s2017/index.html
- *Multi-resolution Ocean Rendering in* Crest Ocean System, Advances in Real-Time Rendering in Games, ACM SIGGRAPH 2019 courses http://advances.realtimerendering.com/s2019/index.htm

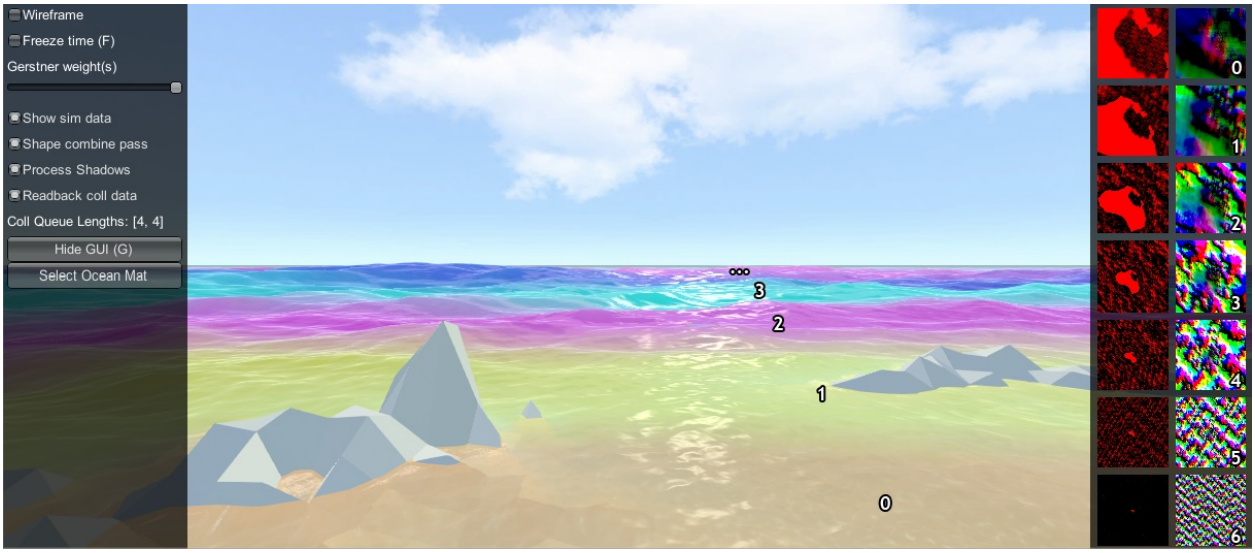
20.1 Core Data Structure

The backbone of *Crest* is an efficient Level Of Detail (LOD) representation for data that drives the rendering, such as surface shape/displacements, foam values, shadowing data, water depth, and others. This data is stored in a multi-resolution format, namely cascaded textures that are centered at the viewer. This data is generated and then sampled when the ocean surface geometry is rendered. This is all done on the GPU using a command buffer constructed each frame by *BuildCommandBuffer*.

Let's study one of the LOD data types in more detail. The surface shape is generated by the Animated Waves LOD Data, which maintains a set of *displacement textures* which describe the surface shape. A top down view of these textures laid out in the world looks as follows:

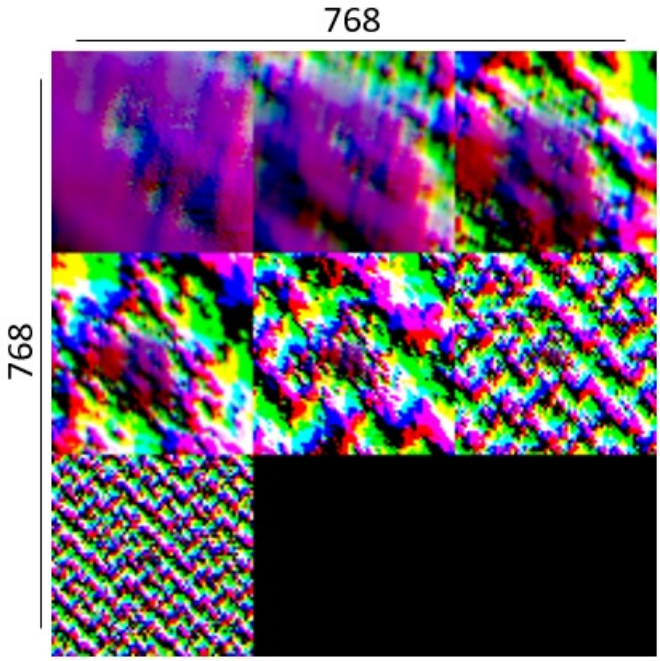


Each LOD is the same resolution (256x256 here), configured on the *OceanRenderer* script. In this example the largest LOD covers a large area (4km squared), and the most detail LOD provides plenty of resolution close to the viewer. These textures are visualised in the Debug GUI on the right hand side of the screen:



In the above screenshot the foam data is also visualised (red textures), and the scale of each LOD is clearly visible by looking at the data contained within. In the rendering each LOD is given a false colour which shows how the LODs are arranged around the viewer and how they are scaled. Notice also the smooth blend between LODs - LOD data is always interpolated using this blend factor so that there are never pops are hard edges between different resolutions.

In this example the LODs cover a large area in the world with a very modest amount of data. To put this in perspective, the entire LOD chain in this case could be packed into a small texel area:



A final feature of the LOD system is that the LODs change scale with the viewpoint. From an elevated perspective, horizontal range is more important than fine wave details, and the opposite is true when near the surface. The *OceanRenderer* has min and max scale settings to set limits on this dynamic range.

When rendering the ocean, the various LOD data are sample for each vert and the vert is displaced. This means that the data is carried with the waves away from its rest position. For some data like wave foam this is fine and desirable.

For other data such as the depth to the ocean floor, this is not a quantity that should move around with the waves and this can currently cause issues, such as shallow water appearing to move with the waves as in #96.

20.2 Implementation Notes

On startup, the *OceanRenderer* script initialises the ocean system and asks the *OceanBuilder* script to build the ocean surface. As can be seen by inspecting the ocean at run-time, the surface is composed of concentric rings of geometry tiles. Each ring is given a different power of 2 scale.

At run-time, the ocean system updates its state in *LateUpdate*, after game state update and animation, etc. *OceanRenderer* updates before other scripts and first calculates a position and scale for the ocean. The ocean GameObject is placed at sea level under the viewer. A horizontal scale is computed for the ocean based on the viewer height, as well as a *_viewerAltitudeLevelAlpha* that captures where the camera is between the current scale and the next scale ($\times 2$), and allows a smooth transition between scales to be achieved.

Next any active ocean data are updated, such as animated waves, simulated foam, simulated waves, etc. The data can be visualised on screen if the *OceanDebugGUI* script from the example content is present in the scene, and if the *Show shape data* on screen toggle is enabled. As one of the ocean data types, the ocean shape is generated using an FFT and copied into the animated waves data. Each wave component is rendered into the shape LOD that is appropriate for the wavelength, to prevent over- or under- sampling and maximize efficiency. A final pass combines the shape results from the different FFT components together. Disable the *Shape combine pass* option on the *OceanDebugGUI* to see the shape contents before this pass.

Finally *BuildCommandBuffer* constructs a command buffer to execute the ocean update on the GPU early in the frame before the graphics queue starts. See the *BuildCommandBuffer* code for the update scheduling and logic.

The ocean geometry is rendered by Unity as part of the graphics queue, and uses the *Crest/Ocean* shader. The vertex shader snaps the verts to grid positions to make them stable. It then computes a *lodAlpha* which starts at 0 for the inside of the LOD and becomes 1 at the outer edge. It is computed from taxicab distance as noted in the course. This value is used to drive the vertex layout transition, to enable a seamless match between the two. The vertex shader then samples any required ocean data for the current and next LOD scales and uses *lodAlpha* to interpolate them for a smooth transition across displacement textures. Finally, it passes the LOD geometry scale and *lodAlpha* to the ocean fragment shader.

The fragment shader samples normal and foam maps at 2 different scales, both proportional to the current and next LOD scales, and then interpolates the result using *lodAlpha* for a smooth transition. It combines the normal map with surface normals computed directly from the displacement texture.

CHAPTER

TWENTYONE

RENDERING NOTES

21.1 Transparency

Crest is rendered in a standard way for water shaders - in the transparent pass and refracts the scene. The refraction is implemented by sampling the camera's colour texture which has opaque surfaces only. It writes to the depth buffer during rendering to ensure overlapping waves are sorted correctly to the camera. The rendering of other transparent objects depends on the case, see headings below. Knowledge of render pipeline features, rendering order and shaders is required to solving incompatibilities.

21.1.1 Transparent Object In Front Of Ocean Surface

Normal transparent shaders should blend correctly in front of the water surface. However this will not work correctly for refractive objects. *Crest* will not be available in the camera's colour texture when other refractive objects sample from it, as the camera colour texture will only contain opaque surfaces. The end result is *Crest* not being visible behind the refractive object.

21.1.2 Transparent Object Behind The Ocean Surface

Alpha blend and refractive shaders will not render behind the water surface. Other transparent objects will not be part of the camera's colour texture when *Crest* samples from it. The end result is transparent objects not being visible behind *Crest*.

On the other hand, alpha test / alpha cutout shaders are effectively opaque from a rendering point of view and may be usable in some scenarios.

21.1.3 Transparent Object Underwater

Note: See the *Underwater Shader API* (in preview) for a more accurate solution.

This is tricky because the underwater effect uses the opaque scene depths in order to render the water fog, which will not include transparents.

The following only applies to the *Underwater Renderer*.

Transparents will need to be rendered after the underwater effect. The underwater effect is rendered at the *BeforeRenderingPostProcessing* event. They can be rendered after the underwater effect using the Render Objects Render Feature set to *BeforeRenderingPostProcessing*. Transparents rendered after the underwater effect will not have the underwater water fog shading applied to them. The effect of the fog either needs to be faked by simply ramping the opacity down to

0 based on distance from the camera, or the water fog shader code needs to be included and called from the transparent shader.

21.2 Render Order

A typical render order for a frame is the following:

- Opaque geometry is rendered, writes to opaque depth buffer.
- Sky is rendered, probably at zfar with depth test enabled so it only renders outside the opaque surfaces.
- Frame colours and depth are copied out for use later in postprocessing.
- Ocean renders early in the transparent queue (queue = 2510).
  - Queue = Geometry+510 *BIRP*. Queue = Transparent-100 *URP*.
  - It samples the postprocessing colours and depths, to do refraction.
  - It reads and writes from the frame depth buffer, to ensure waves are sorted correctly.
  - It stomps over sky - sky is at zfar and will be fully fogged/obscured by the water volume.
- Particles and alpha render. If they have depth test enabled, they will clip against the surface.
- Postprocessing runs with the postprocessing depth and colours.
  - If enabled, underwater postprocess constructs a screenspace mask for the ocean and uses it to draw the underwater effect over the screen.

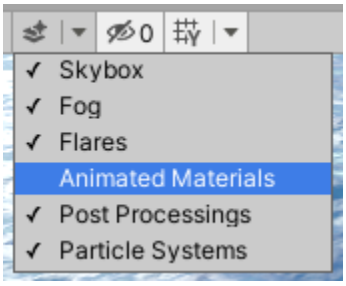
CHAPTER

TWENTYTWO

FREQUENTLY ASKED QUESTIONS

Why does the ocean not update smoothly in edit mode?

By default, the update speed is intentionally throttled by Unity to save power when in edit mode. To enable real-time update, enable *Animated Materials* in the Scene View toggles:



See the Unity Documentation for more information.

I am seeing "Crest does not support OpenGL/WebGL backends." in the editor

It is likely Unity has defaulted to using OpenGL on your platform. You will need to switch to a supported graphics API like Vulkan. You will need to make Vulkan the default by overriding the graphics APIs.

Why aren't my prefab mode edits not reflected in the scene view?

Crest does not support running in prefab mode which means dirty state in prefab mode will not be reflected in the scene view. Save the prefab to see the changes.

Is *Crest* well suited for medium-to-low powered mobile devices?

Crest is built to be performant by design and has numerous quality/performance levers. However it is also built to be very flexible and powerful and as such can not compete with a minimal, mobile-centric water solution such as the one in the *Boat Attack* project. Therefore we target *Crest* at PC/console platforms.

That being said, developers have had success with *Crest* on lower powered platforms like the *Nintendo Switch*. *Apple* devices are good targets as their processors are quite capable all round. *Meta Quest 2* is one device where it will be a struggle to take advantage of *Crest*.

Which platforms does *Crest* support?

Testing occurs primarily on MacOS and Windows. iOS and Android are also periodically tested.

Firstly, make sure your target platform adheres to the *Requirements*.

We have users targeting the following platforms:

- Windows
- MacOS
- Linux *
- PlayStation *
- Xbox *
- Switch * **
- iOS **
- Android/Quest **

* We do not have access to these platforms ourselves.
** Performance is a challenge on these platforms. Please see the previous question.

Crest also supports VR/XR Multi-Pass and Single Pass Instanced rendering.

For additional platform notes, see Platform Support.

Is *Crest* well suited for localised bodies of water such as lakes?

Yes, see *Oceans, Rivers and Lakes* for documentation.

Can *Crest* work with multiplayer?

Yes, the animated waves are deterministic and can be synchronised across the network. For more information see *Network Synchronisation*.

Note however that the dynamic wave sim is not synchronized over the network and should not be relied upon in networked situations.

Errors are present in the log that report *Kernel 'xxx.yyy' not found*

Unity sometimes gets confused and needs assets reimported. This can be done by clicking the *Crest* root folder in the Project window and clicking *Reimport*. Alternatively the *Library* folder can be removed from the project root which will force all assets to reimport.

Can I push the ocean below the terrain?

Yes, this is demonstrated in Fig. 8.1.

Does *Crest* support multiple viewpoints?

Currently only a single ocean instance can be created, and only one viewpoint is supported at a time. We hope to support multiple simultaneous views in the future.

Can I sample the water height at a position from C#?

Yes, see usages of *SampleHeightHelper* class in *SamplingHelpers.cs*. The *OceanRenderer* uses this helper to get the height of the viewer above the water, and makes this viewer height available via the *ViewerHeightAboveWater* property.

Can I trigger something when an object is above or under the ocean surface without any scripting knowledge?

Yes. Please see *Detecting Above or Below Water*.

Does Crest support orthographic projection?

Yes. Please see *Orthographic Projection*.

How do I disable underwater fog rendering in the scene view?

You can enable/disable rendering in the scene view by toggling fog in the scene view control bar.

Can the density of the fog in the water be reduced?

The density of the fog underwater can be controlled using the *Fog Density* parameter on the ocean material. This applies to both above water and underwater. The *Depth Fog Density Factor* on the *Underwater Renderer* can reduce the density of the fog for the underwater effect.

Does Crest support third party weather assets?

Several weather assets provide integrations with Crest. Some will list support in their description, but often they only list support in their documentation.

These may require some code to be inserted into the ocean shader. There is a comment referring to this, search Ocean.shader for 'Azure'.

Please see the Community Contributions section in our Wiki for some integration solutions.

Can I remove water from inside my boat?

Yes, this is referred to as 'clipping' and is covered in section *Clip Surface*.

How to implement a swimming character?

As far as we know, existing character controller assets which support swimming do not support waves (they require a volume for the water or physics mesh for the water surface). We have an efficient API to provide water heights, which the character controller could use instead of a physics volume. Please request support for custom water height providers to your favourite character controller asset dev.

Can I render transparent objects underwater?

See *Transparent Object Underwater*.

Can I render transparent objects in front of water?

See *Transparent Object In Front Of Ocean Surface*.

Can I render transparent objects behind the ocean surface?

See *Transparent Object Behind The Ocean Surface*.

How can I fix the water rendering over the top of HDRP's volumetric clouds when viewing from above?

When viewing volumetric clouds from above, the water may render over them.

For Unity 6 and above, set the *Refraction Model* on the water material to something other than *None* (like *Planar*). This has an overhead so it is recommended to set it to *None* if not required.

For earlier Unity versions, the only way is to set *Surface Type* to *Opaque*. This can be done. The water will look difference in shallow areas so it is advise to set this only when needed.